



Ti₃C₂T_x MXene — Multilayer Powder

Research-grade two-dimensional transition-metal carbide powder

Ti₃C₂T_x MXene is supplied as a multilayer powder derived from Ti₃AlC₂ using a LiF/HCl-derived etching route. The material has mixed -O, -OH, -F and -Cl surface terminations. This grade is supplied for electrochemical, electrocatalysis, conductive-composite and interface research.

KEY FEATURES

- Multilayer; multilayer powder
- Typical conductivity of 1000 ± 500 S/cm
- Mixed -O, -OH, -F, -Cl surface terminations

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal-sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

QUALITY / DOCUMENTATION

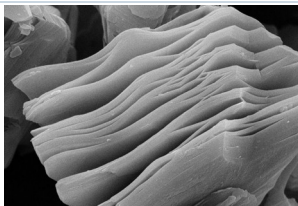
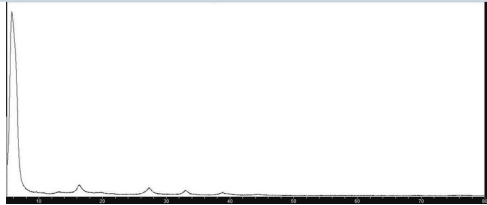
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	Ti ₃ C ₂ T _x MXene
Precursor	Ti ₃ AlC ₂
Product form	Multilayer powder
Preparation	LiF/HCl-derived etching route
Layer structure / count	Multilayer
Typical conductivity	1000 ± 500 S/cm
Surface terminations	-O, -OH, -F, -Cl
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION

	Multilayer Ti ₃ C ₂ T _x — SEM. Representative characterization; not lot-specific.
	Multilayer Ti ₃ C ₂ T _x — XRD. Representative characterization; not lot-specific. Diffraction trace and axes unchanged.

Product positioning: Supplied and technically supported by WiniGen Materials for research and development use.

WINIGEN MATERIALS
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Ti3C2Tx-ML-TDS-001

Representative technical values. Properties can vary with synthesis and lot. Lot-specific documentation and additional analytical details are available subject to review.